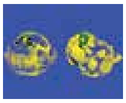

Transformer on the moon
Japan is planning to send a rover to the moon which can transform itself to drive across the surface. **http://dgit.in/jun21-53**

Chinese rover lands on Mars
The Zhurong rover landed on Mars and sent back images, making China the second country to do so. **http://dgit.in/jun21-54**

Illustration of a comet from the Oort cloud
Credit: NASA/SOFIA/Lynette Cook

System Formation

Probing the skies for the clues on how stars and planets are born

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In theory, star systems are supposed to start off as clouds of plasma and dust about a light year across. Most of the matter in this bubble concentrates and collapses to the center into a star. The rest of the matter gets flattened into a disc because of the rotation of the disc around the star. Protoplanets form in these discs, which can be a chaotic time of epic collisions. Large planetary bodies tend to clear their orbits, allowing the planets to settle in a more or less flat plane. More hints about this chaotic time in the early Solar System have been uncovered by a team of researchers that used geochemistry techniques to investigate geological processes... on rocks that had not been subjected to any asteroids. The researchers looked at the minerals formed in the asteroids, and tried to reverse engineer the temperatures at which they had been formed, the conditions that could have created those temperatures, and formulated a rubble pile hypothesis. The early Solar System was a bit like that game of Asteroids, chain reactions of hot bodies smashing into smaller and smaller bits. According to the rubble pile hypothesis, the debris in the accretion disc went through an intense stage of collisions and pulverisations, till they became small, cooled clumps, that coalesced into larger bodies, before commencing into a second round of cooling. After the planets have formed and most of the gases are either concentrated or disappitated, there is a disk of debris left over from the planet formation period, similar to the Kuiper belt in our own Solar System. In the fringes of the system, the objects have more inclined orbits, up to 30 degrees, and the system is less flat.

Astronomers are only beginning to characterise exoplanets, and the data from these reveal that not all stars and planets form in the same way, and that not all planetary systems are flat. For example, we only have gas giants in the outer Solar System, but planet hunting instruments have found hot Jupiters,